# Jia Liu

Ph.D. Student · Division of Emerging Interdisciplinary Areas (EMIA), The Hong Kong University of Science and Technology

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Jia Liu (刘佳)

LOCATION
Hong Kong SAR, China

## RESEARCH INTERESTS

**Multiscale heat transfer · Phonon and electron–phonon Boltzmann transport · Kinetic theory and rarefied gas dynamics · Fast iterative and asymptotic-preserving solvers · Reduced-order modeling**

I develop fast, robust numerical methods for kinetic transport equations, targeting multiscale heat transfer across the ballistic-to-diffusive regimes. My first-authored work includes general synthetic iterative schemes (GSIS) that reduce iteration counts by up to three orders of magnitude for the phonon Boltzmann equation and extend to coupled electron–phonon transport; an asymptotic-preserving solver that models frequency-domain thermoreflectance experiments; and a proper generalized decomposition framework with synthetic preconditioning for parametric phonon transport. Going forward, I aim to build predictive, high-performance transport solvers for thermal management in micro- and nanoscale devices, and to couple them with reduced-order and data-driven models for design and inverse problems.

**Highlights:** 3 first-author articles in JCR Q1 journals (J. Comput. Phys., Commun. Comput. Phys., Int. J. Heat Mass Transfer) · 2 recent preprints · 3 oral presentations at international conferences · SUSTech Outstanding Master's Thesis

## EDUCATION

**Ph.D., Individualized Interdisciplinary Program** — Sep 2024 – Jun 2028 (expected)
Division of Emerging Interdisciplinary Areas (EMIA), The Hong Kong University of Science and Technology
**Dissertation (tentative):** "General Synthetic Iterative Scheme and Proper Generalized Decomposition for Efficient Boltzmann Transport Solvers"
**Advisors:** Prof. Wei Su; Prof. Yanguang Zhou (co-advisor) · **CGA:** 3.80 / 4.30
**Selected coursework:** Advanced Numerical Methods I (A+) and II (A) · Mathematical Methods in Science and Engineering I (A–) · Statistical Machine Learning (A–)

**M.Sc. in Mechanics** — Sep 2021 – Jun 2024
Southern University of Science and Technology (SUSTech), Shenzhen, China
**Thesis:** "General Synthetic Iterative Scheme for Phonon Boltzmann Equation" **(SUSTech Outstanding Master's Thesis)** · **Advisor:** Prof. Lei Wu

**B.Sc. in Information and Computational Science** — Sep 2017 – Jun 2021
Xiangtan University, Xiangtan, China
**GPA:** 3.69 / 4.0 · **Rank:** 3 / 116 · First Class Scholarship (four times)

## RESEARCH EXPERIENCE

**Ph.D. Research · HKUST** Sep 2024 – Present

Multiscale phonon and electron transport · Advisors: Prof. Wei Su, Prof. Yanguang Zhou (co-advisor)

- Developed a general synthetic iterative scheme (GSIS) for fast solution of the coupled electron–phonon Boltzmann equations [J3].
- Co-developed a synthetic iterative scheme for the non-gray phonon Boltzmann transport equation with dual relaxation times [P1].
- Combined proper generalized decomposition (PGD) with synthetic preconditioning for multiscale parametric phonon transport [P2].

**M.Sc. Research · SUSTech** Sep 2021 – Jun 2024

Kinetic theory and multiscale heat transfer · Advisor: Prof. Lei Wu

- Proposed a GSIS for the phonon Boltzmann equation with dual relaxation times, reducing iteration counts by up to three orders of magnitude relative to the conventional iterative scheme [J1].
- Developed a fast-converging, asymptotic-preserving method for simulating frequency-domain thermoreflectance (FDTR) experiments [J2].
- SUSTech Climbing Plan (2022): helped design acceleration algorithms and implement, test and optimize a high-performance solver for multiscale rarefied gas flows.

**Undergraduate Research · Xiangtan University** 2019

National College Students' Innovation and Entrepreneurship Training Program

- Studied the kinetic Boltzmann equation and numerical methods for a computable model of magnetic-field fluidized bed problems.

## PUBLICATIONS

† corresponding author

### Journal Articles

J3 **J. Liu**, X. Xiang, Y. Zhou, W. Su. (2026). General synthetic iterative scheme for fast solving coupled electron–phonon Boltzmann equations. *International Journal of Heat and Mass Transfer*, 260, 128510.

J2 **J. Liu**, L. Wu†. (2023). Fast-converging and asymptotic-preserving simulation of frequency domain thermoreflectance. *Communications in Computational Physics*, 34(1), 65–93.

J1 **J. Liu**, C. Zhang, H. Yuan, W. Su†, L. Wu†. (2022). A fast-converging scheme for the phonon Boltzmann equation with dual relaxation times. *Journal of Computational Physics*, 467, 111436.

### Preprints

P2 **J. Liu**, W. Su. (2026). Proper generalized decomposition with synthetic preconditioning for multiscale parametric phonon transport. arXiv:2610.27880.

P1 D. Shen, **J. Liu**, W. Su. (2026). A synthetic iterative scheme for non-gray phonon Boltzmann transport equation with dual relaxation times. arXiv:2608.28101.

## CONFERENCE PRESENTATIONS

The 1st International Symposium on Non-Equilibrium Transport Phenomena (NETP). Apr 2025

Oral presentation

The 32nd International Symposium on Rarefied Gas Dynamics. Oral presentation — Jul 2022

The 8th Asian Symposium on Computational Heat Transfer and Fluid Flow. Oral presentation — Sep 2021

## HONORS & AWARDS

**Outstanding Master's Thesis Award**, SUSTech — 2024

**Outstanding Graduate (university level)**, SUSTech — 2024

**Outstanding Graduate Teaching Assistant**, SUSTech — 2022

**First Class Scholarship**, Xiangtan University (top 5%, four times) — 2017 – 2021

**Merit Student**, Xiangtan University (top 5%, four times) — 2017 – 2021

**Second Prize (Hunan Division)**, National Undergraduate Mathematical Contest in Modeling — 2019

## TEACHING

**Teaching Assistant · HKUST**
Two courses during Ph.D. study · class size: 30 students

**Teaching Assistant · SUSTech** — Mar 2022 – Jul 2022
2-credit postgraduate course, 62 students · Awarded Outstanding Graduate Teaching Assistant

## SKILLS & LANGUAGES

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|---|---|
| **Numerical methods** | Synthetic iterative acceleration (GSIS) and preconditioning; asymptotic-preserving schemes; reduced-order modeling (proper generalized decomposition); finite volume (high-order WENO), finite difference and finite element methods |
| **Transport physics** | Phonon and electron–phonon Boltzmann transport; kinetic theory and rarefied gas dynamics; thermoreflectance (FDTR) modeling |
| **Scientific computing** | Development, verification and performance optimization of kinetic solvers in C++ and Fortran; MATLAB; LaTeX |
| **Mathematics & data** | Graduate training in advanced numerical analysis, mathematical methods for science and engineering, and statistical machine learning |
| **Languages** | Chinese (native) · English (fluent for research; English-medium Ph.D. program; oral presentations at international conferences) |

## REFERENCES

**Prof. Wei Su**
EMIA
HKUST
Ph.D. advisor

**Prof. Yanguang Zhou**
HKUST
Ph.D. co-advisor

**Prof. Lei Wu**
SUSTech
M.Sc. advisor

Contact details available upon request.